# RetailPulse Analytics Dashboard

Final Internship Project Report Developed by: Syed Moin S Project Title: RetailPulse: Interactive Retail Sales Analytics Dashboard using Python and Streamlit

# Abstract

RetailPulse is a retail data analytics dashboard designed to transform raw sales records into meaningful business insights. The project analyzes historical retail data, customer behavior, product performance, sales trends, and regional performance through interactive visualizations.

# Introduction

Modern businesses generate large amounts of sales data. Manual analysis of this information is difficult and time consuming. RetailPulse provides an easy-to-use dashboard that helps users understand sales patterns and make data-driven decisions.

# Problem Statement

Retail organizations require an efficient way to monitor sales performance, understand customer segments, identify popular products, and analyze trends over time. This project addresses these challenges using data analytics and visualization.

# Objectives

• Analyze retail sales data. • Identify top-selling products and customers. • Compare category and regional performance. • Study monthly and yearly sales trends. • Provide interactive filtering for better analysis.

# Tools and Technologies

Python, Pandas, Streamlit, Jupyter Notebook, VS Code, and GitHub were used for development and implementation of the project.

# Dataset Description

The dataset contains 9,986 records and 18 columns. It includes order information, customer details, product information, categories, regions, and sales values.

# Implementation and Dashboard Modules

The dashboard includes: - Dataset preview and information. - Missing value analysis. - Sales KPIs including total sales, orders, and customers. - Category-wise and region-wise sales analysis. - Top selling products and top customers. - Customer segmentation analysis. - Sub-category analysis. - Monthly and yearly sales trend analysis. - Interactive filters based on region and category.

# Results and Insights

The developed dashboard successfully converts raw retail data into clear visual insights. Users can dynamically filter data and understand different sales patterns, customer behavior, and product performance.

## Dataset Limitation

The provided dataset does not contain a Profit column. Therefore, profit analysis was not included in this project. The analysis focuses on the available sales, customer, product, category, and regional information.

## Conclusion and Future Scope

RetailPulse demonstrates the practical use of data analytics for business intelligence. Future improvements can include adding profit analysis with suitable datasets, database connectivity, advanced machine learning models, and cloud deployment.